

Teleco Customer Churn Analysis

1. Business Overview

This project analyzes customer churn to understand **which customer characteristics, service offerings, contract types, and payment behaviors are associated with customers leaving the company.**

The dataset contains **7,043 customer records and 21 variables**, covering customer demographics, tenure, subscribed services, contract information, billing details, payment methods, and churn status. Before performing the analysis, the data was inspected and prepared for analysis. The TotalCharges field was converted from text to a numerical format, blank values were replaced with zero, and the dataset was checked for missing values and duplicate records. The analysis found **no missing values and no duplicate customer records.**

The dataset represents customers with an average tenure of approximately **32.4 months**, average monthly charges of approximately **\$64.76**, and average total charges of approximately **\$2,279.73**. Tenure ranges from **0 to 72 months**, while monthly charges range from approximately **\$18.25 to \$118.75**.

2. Overall Churn Situation

The most important finding from the analysis is that **26.54% of customers have churned**, while approximately **73.46% have remained with the company**. In other words, roughly **1 out of every 4 customers** in the dataset has left the company.

This indicates that customer retention is a significant business consideration. Rather than treating all customers equally, the analysis suggests that churn is concentrated within particular customer groups.

The subsequent visualizations were therefore used to identify the characteristics and services most strongly associated with customer churn.

Customer Churn Insights

3.1 Contract Type Is a Major Churn Indicator

Contract type is one of the clearest patterns identified in the analysis.

Customers on **month-to-month contracts show substantially higher churn**, whereas customers with **one-year and two-year contracts are much more likely to remain with the company**. The analysis therefore indicates a strong relationship between contract commitment and customer retention.

The churn rates by contract type are approximately:

Contract Type	Approx. Churn Rate
Month-to-month	42.7%
One year	11.3%
Two year	2.8%

This is a significant difference. Customers on month-to-month contracts churn at a rate several times higher than customers on longer-term contracts.

Business implication

The company should consider strategies that encourage month-to-month customers to move toward longer-term contracts. Possible approaches include:

- Contract-upgrade incentives
- Discounts for one-year or two-year commitments
- Loyalty benefits for longer contracts
- Personalized offers for high-risk month-to-month customers

The objective should not simply be to push customers into longer contracts, but to **increase perceived value and commitment while reducing the likelihood of early customer departure**.

3.2 Tenure Shows a Strong Relationship with Churn

Customer tenure is another important factor.

The tenure visualization shows that customers with **very short relationships with the company are considerably more likely to churn**, particularly customers who have been using the service for only **one or two months**. Customers who have remained with the company for longer periods show much stronger retention.

The dataset's median tenure is **29 months**, while the average tenure is approximately **32.37 months**.

This creates a clear customer-lifecycle pattern:

New customer → Higher churn risk → Longer tenure → Greater retention

Business implication

The first few months of the customer relationship appear to be a particularly important retention window.

The company could therefore focus on:

- Stronger onboarding during the first 90 days
- Early customer satisfaction checks
- Proactive technical support
- New-customer incentives
- Personalized communication after installation or activation
- Early identification of customers showing signs of dissatisfaction

The key insight is that **retention efforts should begin early rather than waiting until a customer is already considering cancellation.**

4. Senior Citizen Customers Show Higher Churn

The analysis also compares churn between senior citizens and non-senior citizens.

Senior citizens represent approximately **16.2% of the customer base**, based on the descriptive statistics in the notebook.

The churn comparison shows that senior-citizen customers have a **higher churn proportion** than non-senior customers. The approximate churn rates are:

Customer Segment	Approx. Churn Rate
Non-Senior Citizen	23.6%
Senior Citizen	41.7%

Therefore, although senior citizens form a smaller portion of the overall customer base, their churn rate is considerably higher.

Business implication

This segment may benefit from more targeted customer support and engagement. In particular, the company could investigate:

- Ease of service usage
- Technical-support requirements
- Pricing sensitivity
- Customer-service experience

- Understanding of available services and benefits

However, the analysis identifies the **relationship with churn**, rather than proving why senior customers churn. Further investigation would be required to identify the underlying cause.

5. Internet Service Type Has a Strong Relationship with Churn

Internet service is another major area highlighted by the analysis.

The visualization indicates that **fiber-optic customers churn more frequently than DSL customers and customers without internet service.**

Approximate churn rates are:

Internet Service	Approx. Churn Rate
Fiber optic	41.9%
DSL	19.0%
No internet service	7.4%

The difference is substantial. Fiber-optic customers have a churn rate of roughly **42%**, compared with approximately **19% for DSL customers.**

Business implication

This is an important area for further investigation.

The analysis itself suggests that the higher churn among fiber customers **may indicate dissatisfaction with service quality or pricing.** However, the current analysis cannot determine which of these factors is responsible.

Therefore, management should investigate:

- Fiber pricing compared with competitors
- Customer perception of value
- Service reliability
- Technical issues
- Installation experience
- Customer-support interactions
- Whether fiber customers are disproportionately concentrated among month-to-month contracts

This could potentially uncover an important business problem because fiber customers appear to represent a **high-risk customer segment**.

6. Online Security Is Associated with Better Retention

The analysis of additional services shows an important pattern for **Online Security**.

Customers **without Online Security** show higher churn compared with customers who subscribe to the service. The notebook specifically identifies Online Security as a potential **retention lever**.

Approximate churn rates are:

Online Security	Approx. Churn Rate
No	29.1%
Yes	14.6%

Customers without Online Security therefore show roughly **twice the churn rate** of customers who have the service.

Business implication

Online Security could potentially be incorporated into retention-oriented packages.

For example, customers at high churn risk could receive:

- Discounted security packages
- Free trials
- Bundled security and internet plans
- Personalized recommendations based on their existing services

Importantly, the analysis demonstrates an **association**, not proof that subscribing to Online Security directly prevents churn.

7. Tech Support Is Another Important Retention Factor

A similar pattern appears for **Tech Support**.

Customers who do not have Tech Support show considerably higher churn than customers who subscribe to it. Your analysis specifically highlights Tech Support as one of the services that could act as a retention lever.

The approximate churn rates are:

Tech Support	Approx. Churn Rate
No	31.2%
Yes	15.2%

This suggests that customers with access to technical support are substantially more likely to stay.

Business implication

The company could consider:

- Proactively offering technical support to new customers
- Bundling Tech Support with internet plans
- Providing temporary free support for high-risk customers
- Improving awareness of existing support services

This may be especially relevant for customers using fiber-optic internet, where the analysis already shows elevated churn.

8. Online Backup and Device Protection Are Also Associated with Churn

The analysis identifies **Online Backup** and **Device Protection** as additional services where customers without these services show higher churn.

This suggests that customers who use more protective or support-oriented services may demonstrate stronger retention.

The pattern can be summarized as:

More support/security services → Lower observed churn

However, as with Online Security and Tech Support, this should be interpreted as an association rather than direct causation.

Business implication

The company could explore bundled packages combining:

Internet + Security + Backup + Device Protection + Technical Support

Such packages may increase perceived customer value while simultaneously creating a stronger relationship between the customer and the company.

9. Streaming Services Have a Weaker Relationship with Churn

The analysis also examines **Streaming TV** and **Streaming Movies**.

Compared with Online Security, Tech Support, Online Backup, and Device Protection, these entertainment services do not demonstrate as strong a relationship with churn in the analysis.

This indicates that entertainment add-ons may not be the primary drivers of customer retention.

Business implication

Retention strategies should therefore prioritize **security, support, service experience, and contract structure** over relying primarily on entertainment features.

Streaming services may still contribute to customer value, but based on this analysis they should not be considered the strongest churn-reduction lever.

10. Phone Service and Multiple Lines Have Relatively Limited Impact

The analysis also evaluates **Phone Service** and **Multiple Lines**.

Compared with internet service, contract type, tenure, and support/security services, these variables show a relatively weaker relationship with churn.

This suggests that these services should not be the primary focus of a churn-reduction strategy.

From a business perspective, resources should instead be directed toward the variables showing stronger differences in churn behavior.

11. Payment Method Is an Important Churn Pattern

Payment method is another notable finding.

Customers using **Electronic Check** show higher churn compared with customers using other payment methods such as bank transfer, credit card, and mailed check. Your analysis suggests that electronic-check customers may represent a more volatile customer segment.

Approximate churn rates are:

Payment Method	Approx. Churn Rate
Electronic check	45.3%
Mailed check	19.1%
Bank transfer (automatic)	16.7%
Credit card (automatic)	15.2%

The difference is particularly notable for electronic checks: the churn rate is approximately **45%**, meaning nearly **one in two customers** in this segment has churned.

Business implication

This segment should receive additional attention.

The company could investigate:

- Why electronic-check customers are more likely to leave
- Whether this group overlaps heavily with month-to-month customers
- Whether payment friction contributes to cancellation
- Whether automatic payment incentives could improve retention

The company could also consider encouraging customers to use more stable payment methods through incentives or simplified payment enrollment.

12. Paperless Billing Shows a Relationship with Higher Churn

The analysis also identifies a relationship between **Paperless Billing** and the higher-churn electronic-check customer group.

This does not necessarily mean that paperless billing itself causes customers to churn. Instead, it may reflect the characteristics of the customers who use this billing and payment combination.

Therefore, paperless billing should be viewed as a **customer-profile indicator** rather than automatically treated as a cause of churn.

13. Gender Does Not Appear to Be a Significant Churn Driver

The gender analysis shows relatively similar churn patterns between male and female customers.

Your notebook concludes that **gender has no meaningful effect on customer churn**.

This is important from a business perspective because it suggests that gender-based segmentation would not be a priority for churn reduction.

Instead, the company should focus on behavioral and service-related variables such as:

- Contract type
- Tenure
- Internet service
- Support services
- Security services
- Payment method

These variables provide much stronger signals in the analysis.

14. Overall Customer Risk Profile

When the findings are considered together, the analysis suggests that the **highest-risk customer profile** is associated with several characteristics occurring together:

Short-tenure + month-to-month contract + fiber-optic internet + electronic-check payment + limited security/support services

For example, a new customer on a month-to-month contract who uses fiber-optic internet, pays by electronic check, and does not have Online Security or Tech Support would fall into several of the higher-churn categories identified by the analysis.

This does **not** mean that every customer with these characteristics will churn. Rather, the combination provides a useful basis for identifying customers who may deserve additional retention attention.

15. Strategic Recommendations

Based on the patterns identified in the analysis, the following actions could help the company improve customer retention:

1. Focus on month-to-month customers

Month-to-month customers have the highest observed churn rate, approximately **42.7%**.

The company should prioritize this segment with personalized offers and incentives to transition customers toward longer-term contracts.

2. Strengthen early-stage customer retention

Customers with very short tenure show significantly higher churn.

A structured **30-, 60-, and 90-day customer-retention program** could help identify dissatisfaction before customers leave.

3. Investigate fiber-optic customer experience

Fiber customers have an observed churn rate of approximately **41.9%**, substantially above DSL customers.

Management should investigate whether the issue is related to pricing, service quality, technical problems, installation, or customer expectations.

4. Promote security and technical-support services

Customers without Online Security and Tech Support show substantially higher churn.

These services could be incorporated into targeted retention offers or bundled plans.

5. Investigate electronic-check customers

Electronic-check users have an observed churn rate of approximately **45.3%**, making this one of the most notable high-risk segments.

The company should investigate the underlying reason for this behavior and explore whether alternative payment methods can improve customer stability.

6. Use customer segmentation for retention

Rather than using a single retention strategy for everyone, customers should be segmented according to churn risk.

For example:

High-risk customers:

Short tenure + month-to-month + electronic check + fiber + limited support services

Medium-risk customers:

Month-to-month customers with moderate tenure or higher monthly charges

Lower-risk customers:

Long-tenure customers with one- or two-year contracts and multiple support/security services

16. Final Conclusion

The analysis of **7,043 customers** shows an overall churn rate of **26.54%**, meaning that more than one-quarter of the customer base has churned.

The strongest patterns are not related to gender but are instead associated with **customer tenure, contract type, internet service, support/security services, and payment method**.

The most significant findings are:

- **26.54%** of customers have churned.
- **Month-to-month customers have approximately 42.7% churn**, compared with approximately **11.3% for one-year contracts and 2.8% for two-year contracts**.
- **Senior citizens have approximately 41.7% churn**, compared with approximately **23.6% among non-senior customers**.
- **Fiber-optic customers have approximately 41.9% churn**, compared with approximately **19.0% for DSL customers**.
- Customers without **Online Security** show approximately **29.1% churn**, compared with **14.6%** among customers with the service.
- Customers without **Tech Support** show approximately **31.2% churn**, compared with approximately **15.2%** among customers with the service.
- **Electronic-check users have approximately 45.3% churn**, considerably higher than customers using other payment methods.
- **Gender does not show a meaningful relationship with churn** in the analysis.
- Streaming services, Phone Service, and Multiple Lines show comparatively weaker relationships with churn than the major factors above.

Overall, the analysis indicates that **customer churn is primarily associated with the customer's stage in the relationship, level of contractual commitment, internet-service type, support/security adoption, and payment behavior**.

The key business opportunity is therefore to identify high-risk customers early and provide targeted interventions. In particular, **new month-to-month customers, fiber-optic**

customers, electronic-check users, and customers without support/security services should receive greater attention.

By combining these insights into a targeted retention strategy, the company can move from a broad, one-size-fits-all approach toward a **data-driven customer retention program focused on the segments most strongly associated with churn.**